# S. M. Sakeef Sani

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**Research Impact** — **10** Publications · **502** Citations · **h-index: 5** · **i10-index: 3** · *Source: Google Scholar*

## Experience

- **Hinton Bit Corp** Dhaka, Bangladesh
  *Co-founder & Director of Operations* *2026 – Present*
  - **RAVEN — Bangladesh's First National OSINT Platform**: Conceived, led a cross-functional engineering team, and delivered **RAVEN** — Bangladesh's **first open-source intelligence (OSINT) platform**, now in active operational use by national Law Enforcement Agencies. An **8-module** end-to-end intelligence suite actively deployed with national law enforcement agencies, covering: real-time social media monitoring with AI-graded post & sentiment analysis; virality & trend detection; autonomous LLM-powered event watchtower; deepfake & AI-media forensics with manipulation heatmaps; agentic multi-tool claim-origin investigation; face detection/recognition; NID-linked identity dossiers; and 3-hop relationship network mapping.
  - **Enterprise & Government Solutions**: Delivers mission-critical software, data & AI systems, and compute infrastructure (GPU provisioning, server deployment, network hardening) for government and private-sector clients alongside globally renowned technology partners.
  - **Product Portfolio Leadership**: Drives strategy and end-to-end delivery across: **RAVEN** (national security intelligence), **Citizen Portal** (AI-powered complaint & case management for public agencies, with WhatsApp bot intake and immutable audit trails), and **LMSMate** (white-label EdTech platform with live classes, AI quizzes, and local payments).
- **British American Tobacco** Bangladesh
  *Territory Officer* *2025 – 2026*
  - **Distribution Analytics**: Managed a **420 Mn BDT** annual distribution network; designed an innovative rural coverage model for remote 'Char' regions using satellite imagery, geographic data, and ROI analytics.
  - **Shipment Automation Application**: Engineered a real-time automated shipment application for route-to-market efficiency, improving distributor performance tracking across the territory.
- **mHealth Lab, BUET** Dhaka, Bangladesh
  *Research Engineer* · Youngest recruit; joined in sophomore year *2021 – 2025*
  - **mLabLLM** [*Published, AAAI 2025*]: Core contributor to **mLabLLM** — a fine-tuned LLaMA 3.2 model achieving **82.8% Top-3 accuracy** for differential diagnosis of Dengue, Malaria, and Chikungunya via Bayesian reasoning, LoRA, and model pruning. Published at *AAAI 2025*.
  - **Real-time Physiological Data Collector**: Designed and deployed a Raspberry Pi–based real-time physiological data collection system for field deployment in low-resource clinical settings.
  - **3D Landmarks from Sparse 2D Video**: Developed a pipeline to reconstruct 3D anatomical landmarks from sparse 2D video, improving imaging accuracy under low-compute constraints.
- **PakhiDrone BD** Bangladesh
  *Founder & Chief Marketing and Licensing Officer (CMLO)* *2023 – 2024*
  - **Agro-Drone for Precision Farming**: Founded and led development of an agro-drone tailored for Bangladesh's rural smallholder terrain; secured **UIHP** grant and BHTPA incubation support.
- **Enigmatter Studio** Bangladesh
  *Founder, Animator & Lead VFX Artist* *2021 – 2025*
  - **Award-Winning 3D Animation & VFX**: Produced award-winning short films using Blender and Unreal Engine; spanned iOS & Android game development, sci-fi writing, and complex 3D scene design.

## Publications

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- **Efficient Antihallucinogenic AI for Tropical Medicine: A Probabilistic Framework for Differential Diagnosis**: **AAAI 2025** · Fine-tuned LLaMA 3.2 with Bayesian reasoning & LoRA; 82.8% Top-3 accuracy for tropical disease diagnosis.
  *openreview.net/forum?id=E6J9IFPE0w*
- **A Web-Based Mpox Skin Lesion Detection System Considering Racial Diversity**: **Biomedical Signal Processing and Control, Elsevier 2025** · Racially diverse MSLD v2.0 dataset; ResNet50 benchmarking via HAM10000 transfer learning.
  *sciencedirect.com/science/article/pii/S1746809424008000*
- **Monkeypox Skin Lesion Detection Using Deep Learning Models: A Feasibility Study**: **arXiv 2022** · Foundational MSLD dataset; ~82.96% accuracy distinguishing mpox from similar skin conditions.
  *arxiv.org/abs/2207.03342*
- **BUET Multi-Disease Heart Sound Dataset (BMD-HS): A Comprehensive Auscultation Dataset for Computer-Aided Diagnostics**: **arXiv 2024** · 864 multi-label heart sound recordings for complex valvular diseases; bridges auscultation with AI diagnostics.
  *arxiv.org/abs/2409.00724*

## Projects

- **mLabLLM — Efficient Antihallucinogenic AI for Tropical Medicine** (AI, LLM, Bayesian Framework): Fine-tuned LLaMA 3.2 for differential diagnosis of Dengue, Malaria, and Chikungunya; 82.8% Top-3 accuracy via LoRA and model pruning for real-time clinical support. *Python, TensorFlow* (2025)
- **MSLD v2.0 — Racial-Diversity-Aware Mpox Detection System** (Deep Learning, Fairness AI): Created the racially diverse MSLD v2.0 dataset; benchmarked ResNet50 via HAM10000 transfer learning; deployed Flask prototype for inclusive, rapid mpox screening. *Python, Keras, Flask* (2025)
- **BMD-HS — BUET Multi-Disease Heart Sound Dataset** (Dataset Development, AI Diagnostics): Co-developed 864 multi-label heart sound recordings for complex valvular diseases, bridging clinical auscultation with AI diagnostics. *Python, Audio Processing* (2024)
- **FetoSynth — Fetal Heart Sound Localization [2nd Place, Johns Hopkins Global Health Track 2024]**: Real-time acoustic system for fetal heart sound localization and prenatal monitoring in low-resource settings. *Raspberry Pi, AI Models* (2024)
- **DengueDrops — Smart IV Fluid Calculator [2nd Place, Johns Hopkins Digital Health Track 2024]**: AI-powered IV fluid optimization and hospital workflow management for dengue patients. *Mobile Development, AI* (2024)
- **MSLD v1.0 — Monkeypox Skin Lesion Detection Feasibility Study** (Deep Learning, Dataset Curation): Foundational MSLD dataset; evaluated pre-trained models achieving ˜82.96% accuracy distinguishing mpox from similar conditions. *Python, TensorFlow* (2022)

## Education

- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  *Bachelor of Science in Biomedical Engineering* — *2020 – 2025*
  *Courses: Machine Learning, Artificial Intelligence, Deep Learning, Biomedical Imaging, Data Structures* — *CGPA: 3.57/4.00*

## Honors & Awards

- **Honda Young Engineer and Scientist (Y-E-S) Award 2023** — Honda Foundation, Bangladesh
- **2nd Place** — Johns Hopkins Healthcare Design Competition, *Global Health Track* 2024 — Team FetoSynth
- **2nd Place** — Johns Hopkins Healthcare Design Competition, *Digital Health Track* 2024 — Team DengueDrops
- **Innovation Award** — International Planetary Aerial Systems Challenge (IPAS) 2021 — $1^{st}$ in Bangladesh, $8^{th}$ Globally
- **UIHP Winner 2023** — University Innovation Hub Programme, BHTPA — PakhiDrone BD